

Special Relativity Notes

Relativistic Mass, Kinetic Energy and the Mass–Energy–Momentum Relation

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1 Variation of Mass with Velocity

We have to use conservation of momentum in both frames and find how m_1 and m_2 are related to each other with respect to the frame velocity.

Conservation of momentum (frame S)

$$m_1 v_1 + m_2 v_2 = (m_1 + m_2) v$$

Apply the rule of velocity transformation. In frame S' the two masses approach each other with speed u' ; S' itself moves with velocity v relative to S . Then

$$v_1 = \frac{u' + v}{1 + \frac{u'v}{c^2}}, \quad v_2 = \frac{-u' + v}{1 - \frac{u'v}{c^2}}.$$

Let $x = \frac{u'v}{c^2}$. Substituting into the momentum equation:

$$m_1 \left(\frac{u' + v}{1 + x} \right) + m_2 \left(\frac{-u' + v}{1 - x} \right) = (m_1 + m_2) v$$

Group the m_1 and m_2 terms separately:

$$m_1 \left[\frac{u' + v}{1 + x} - v \right] = m_2 \left[v - \frac{v - u'}{1 - x} \right]$$

Simplifying each bracket:

$$\begin{aligned} \frac{u' + v}{1 + x} - v &= \frac{u' + v - v(1 + x)}{1 + x} = \frac{u' - vx}{1 + x} \\ v - \frac{v - u'}{1 - x} &= \frac{v(1 - x) - (v - u')}{1 - x} = \frac{u' - vx}{1 - x} \end{aligned}$$

So

$$m_1 \frac{u' - vx}{1 + x} = m_2 \frac{u' - vx}{1 - x}$$

Cancelling the common factor $(u' - vx)$:

$$\boxed{\frac{m_1}{m_2} = \frac{1 + x}{1 - x}} \quad \left(x = \frac{u'v}{c^2} \right)$$

Relating m_1/m_2 to v_1 and v_2

Using the relativistic velocity-addition identity $1 - \frac{w^2}{c^2} = \frac{\left(1 - \frac{u'^2}{c^2}\right)\left(1 - \frac{v^2}{c^2}\right)}{(1 \pm x)^2}$ for $w = v_{1,2}$:

$$c^2 - v_1^2 = \frac{(c^2 - u'^2)(c^2 - v^2)}{c^2(1+x)^2}, \quad c^2 - v_2^2 = \frac{(c^2 - u'^2)(c^2 - v^2)}{c^2(1-x)^2}$$

Taking the ratio:

$$\frac{c^2 - v_2^2}{c^2 - v_1^2} = \frac{(1+x)^2}{(1-x)^2} \implies \frac{\sqrt{c^2 - v_2^2}}{\sqrt{c^2 - v_1^2}} = \frac{1+x}{1-x} = \frac{m_1}{m_2}$$

$$\implies m_1 \sqrt{c^2 - v_1^2} = m_2 \sqrt{c^2 - v_2^2} \implies m_1 \sqrt{1 - \frac{v_1^2}{c^2}} = m_2 \sqrt{1 - \frac{v_2^2}{c^2}} = \text{constant}$$

This constant is the **rest mass**:

$$m_i \sqrt{1 - \frac{v_i^2}{c^2}} = m_0$$

$$m_i = \frac{m_0}{\sqrt{1 - \frac{v_i^2}{c^2}}}$$

If the object is at rest in the moving frame and the frame itself moves with velocity v :

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} = \gamma m_0$$

where

- m — relativistic mass
- m_0 — rest mass
- v — velocity of the relativistic frame

$$m \geq m_0, \quad T \geq T_0, \quad L \leq L_0$$

2 Variation of Kinetic Energy with Velocity

In the non-relativistic frame,

$$KE = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$

Relativistic kinetic energy. Starting from work done by a force:

$$KE = T = \int_0^x F dx = \int \frac{dp}{dt} v dt = \int d(mv) v$$

With the relativistic mass $m = \gamma m_0$, the momentum is $p = \gamma m_0 v$, so

$$T = \int \frac{d(\gamma m_0 v)}{dt} v dt = m_0 \int_0^v d(\gamma v) v$$

Since $\gamma v = \frac{v}{\sqrt{1 - v^2/c^2}}$, differentiate with respect to v :

$$\begin{aligned} \frac{d(\gamma v)}{dv} &= \left(1 - \frac{v^2}{c^2}\right)^{-1/2} + v \left(-\frac{1}{2}\right) \left(1 - \frac{v^2}{c^2}\right)^{-3/2} \left(-\frac{2v}{c^2}\right) \\ &= \left(1 - \frac{v^2}{c^2}\right)^{-1/2} + \frac{v^2}{c^2} \left(1 - \frac{v^2}{c^2}\right)^{-3/2} \\ &= \left(1 - \frac{v^2}{c^2}\right)^{-3/2} \left[\left(1 - \frac{v^2}{c^2}\right) + \frac{v^2}{c^2}\right] \\ &= \left(1 - \frac{v^2}{c^2}\right)^{-3/2} = \gamma^3 \end{aligned}$$

so $d(\gamma v) = \gamma^3 dv$, and

$$T = m_0 \int_0^v \gamma^3 v dv = m_0 \int_0^v \frac{v}{\left(1 - \frac{v^2}{c^2}\right)^{3/2}} dv$$

Substitution: let $u = 1 - \frac{v^2}{c^2}$, so $du = -\frac{2v}{c^2} dv$, i.e. $v dv = -\frac{c^2}{2} du$:

$$\begin{aligned} T &= -\frac{m_0 c^2}{2} \int u^{-3/2} du = -\frac{m_0 c^2}{2} \left(-2u^{-1/2}\right) = m_0 c^2 u^{-1/2} \\ &= \frac{m_0 c^2}{\sqrt{1 - \frac{v^2}{c^2}}} \bigg|_0^v = m_0 c^2 \left(\frac{1}{\sqrt{1 - v^2/c^2}} - 1\right) \end{aligned}$$

$$\boxed{KE = T = m_0 c^2 (\gamma - 1)} \quad (m = \gamma m_0)$$

$$T = mc^2 - m_0 c^2 \quad \implies \quad mc^2 = T + m_0 c^2$$

$$E = T + m_0 c^2$$

where E is the **total relativistic energy**, T is the kinetic energy, and $m_0 c^2$ is the **rest energy**.

3 Mass–Energy–Momentum Relation

$$E = \frac{m_0 c^2}{\sqrt{1 - v^2/c^2}} = \gamma m_0 c^2, \quad p = \gamma m_0 v, \quad p^2 = \gamma^2 m_0^2 v^2$$

$$\begin{aligned}
p^2 c^2 &= \gamma^2 m_0^2 v^2 c^2 \\
E^2 - p^2 c^2 &= \gamma^2 m_0^2 c^4 - \gamma^2 m_0^2 v^2 c^2 \\
&= \gamma^2 m_0^2 c^2 (c^2 - v^2) \\
&= \gamma^2 m_0^2 c^4 \left(1 - \frac{v^2}{c^2}\right) \\
&= m_0^2 c^4 \quad \left(\because \gamma^2 \left(1 - \frac{v^2}{c^2}\right) = 1\right)
\end{aligned}$$

$$\therefore E^2 - p^2 c^2 = m_0^2 c^4$$

$$E^2 = p^2 c^2 + m_0^2 c^4$$